Adeesha - Practical Projects

Project 1: Customer Churn Prediction System

A production ML system predicting customer churn for a SaaS company with automated retraining pipeline.

Tech Stack: Python, XGBoost, Airflow, MLflow, AWS
Business Problem:
- 15% monthly churn rate costing $500K annually
- No visibility into at-risk customers

Solution:
- Feature engineering from user behavior logs
- XGBoost model with 89% recall
- Automated daily predictions
- Email alerts for high-risk customers

Impact:
- Reduced churn by 22% in 3 months
- Saved $110K in first quarter
- ROI: 5x investment

Client: SaaS Startup (NDA)

Project 2: Inventory Optimization System

An ML-powered inventory management system for retail chain optimizing stock levels and reorder points.

Tech Stack: Python, Prophet, PostgreSQL, Django, Docker
Business Problem:
- $2M annual loss from overstocking/understocking
- Manual ordering based on gut feeling

Solution:
- Time series forecasting for demand prediction
- Safety stock optimization
- Automated reorder recommendations
- Dashboard for inventory managers

Impact:
- Reduced inventory costs by 18%
- Improved stock availability to 98%
- Freed up $400K in working capital

Client: Retail Chain (50 stores)

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Project 3: Document Processing Pipeline

An automated document extraction system for insurance claims processing.

Tech Stack: Python, Tesseract, Transformers, FastAPI, Redis
Business Problem:
- Manual processing of 1000+ claims daily
- Average processing time: 15 minutes per claim
- High error rate in data entry

Solution:
- OCR for document digitization
- NLP for field extraction
- Validation rules engine
- Integration with claims management system

Impact:
- Reduced processing time to 2 minutes per claim
- Error rate reduced by 85%
- Saved 3 FTE equivalent

Client: Insurance Company